Week 6 SI Practice Problems
PSYC 2317 – Spring 2026

1. Central Limit Theorem tells us the shape of the distribution of sample means should be normal when the population is _____ or *n* approaches _____.
   A. Positively skewed, 30
   B. Normal, 30
   C. Normal, +2.0
   D. Unbiased, 30

2. Standard error is the spread of_____.
   A. Sample scores
   B. Sample sizes
   C. Sample variances
   D. Sample means

3. For a normal population with $\mu = 100$ and $\sigma = 12$, what is the probability of obtaining a sample mean greater than 106 for the following sample sizes:
   A. $n = 4$
   B. $n = 16$
   C. $n = 36$
   D. What is probability of obtaining a score less than 106?

4. A population of scores with $\mu = 38$ and $\sigma = 6$ is standardized to create a new population with $\mu = 50$ and $\sigma = 10$. What is the new value for each of the following scores from the original population?
   A. $X = 26$
   B. $X = 35$

5. A set of exam scores are reported as X values and z-scores. On this exam a score of $X = 75$ corresponds to a z-score of $z = +2.00$, and a score of $X = 60$ corresponds to a z-score of $z = -1.00$. What are the values for the mean and standard deviation for this exam?
6. Name the following notations:
   A. μ
   B. σ
   C. M
   D. σ_M
   E. σ^2

7. The average high score in Tetris is 2,000 with a standard deviation of 400. To qualify as a professional Tetris player, you need an average high score of 2102 or higher. If you randomly selected 50 Tetris players and averaged their high scores, what is the probability the group average is above the professional level?

8. As sample size increases, does the standard error increase or decrease? Why?
   A. Increase
   B. Decrease

9. With a population mean of $\mu = 80$ and $\sigma = 13$, you draw 3 samples, shown in the table. Without any computation, answer the following questions:
   A. What is changing from Sample A → B → C?
   B. How will that affect standard error?
   C. How will that affect their z-scores?

SAMPLE	*M*	*n*
A	70	3
B	70	15
C	70	50

10. Compute standard errors and z-scores for the samples in question 10.

11. With a population mean of $\mu = 150$, and $\sigma = 19$, what is the probability of a sample mean falling between 140 and 155 if the sample size is 30? DRAW THE DISTRIBUTION!